LegalLOB: Generating Structurally Valid Level-2 Limit Order Book Trajectories with Diffusion Models

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Abstract

A limit order book (LOB) records the buy and sell orders waiting in an electronic market. Diffusion models can generate plausible future Level-2 (L2) books for simulation and stress testing, but a valid book must obey three rules: the best ask stays above the best bid, prices stay ordered across levels, and displayed depth is non-negative. Existing LOB diffusion models output unconstrained prices and depths, so their samples often break these rules; training penalties only make violations rarer, and exact projection repairs samples from outside the generator. To make validity a property of the generator itself, we propose LegalLOB, a diffusion model in which every decoded book, from each intermediate prediction to the final output, satisfies these rules by construction (as continuous prices and depths). To turn every constraint into a lower bound on one number, it describes each book by a free midpoint and the quantities the rules constrain: the spread, the gaps between adjacent levels, and the depths. Softplus links keep each of these numbers above its bound, and cumulative sums rebuild the price ladders from them, leaving the diffusion objective and sampler unchanged. On four Nasdaq datasets, unconstrained and penalty-trained baselines produce invalid books in 48–100% of snapshots, whereas LegalLOB produces none, and along a 100-step rollout every prediction it decodes is valid without projection, where in-loop repair needs 200 projection calls. It runs at 99.6% of the unconstrained model's speed, and on the densest dataset its probabilistic scores stay close to those of the best repair-based baseline, which still leads on most fidelity metrics. Code: https://github.com/moderncavemann/LegalLOB.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Applied computing** → *Economics*.

Keywords

limit order books, diffusion models, market simulation, constrained generation, structural validity

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1 Introduction

A limit order book (LOB) lists the outstanding buy and sell orders of an electronic market at each price level [20]; its Level-2 (L2) view shows the best K price levels on each side and the displayed depth at each level. Generative LOB models produce plausible future market states for simulation, stress testing, synthetic data, and model validation [5, 16, 22, 26]. Diffusion models suit this task because they learn high-dimensional conditional distributions rather than point forecasts [3, 21, 31, 35, 36]; a future of ten snapshots with ten levels per side already contains 400 coupled prices and depths.

An L2 book, however, is not an arbitrary vector of real numbers. In a real book, the best ask lies above the best bid, ask prices rise and bid prices fall away from the best quotes, and displayed depth is non-negative [20]. Standard diffusion models output unconstrained values, so even a model with low denoising error can output crossed quotes, inverted levels, or negative depth. Such books must be rejected or repaired before risk calculations, downstream models, or the next step of a recursive simulation can use them.

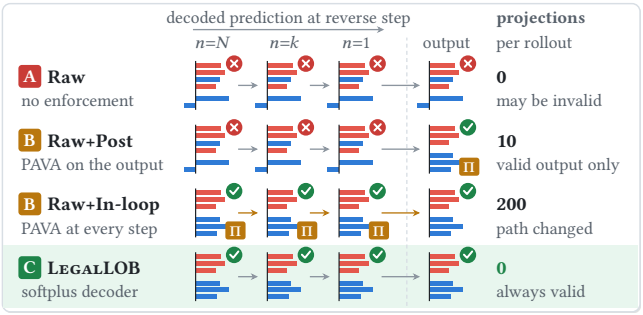


Figure 1: Validity along the sampling path (schematic; ✗: may be invalid; Π: exact projection). Each row shows the decoded clean-state prediction at reverse steps N, k, and 1, the output, and the projection calls per 100-step GOOG-Event rollout (Table 4); in-loop projections feed back into sampling.

Training on valid data alone gives no guarantee, and apart from diffusion models that modify the stochastic dynamics or learn an additional map for general constraint sets (Section 2), existing remedies fall into two paradigms. *Training-time* remedies change what the model learns: a structural penalty [13] makes violations costly

during training but cannot rule them out at sampling time. *Repair-based* remedies project decoded books onto the valid set after or during sampling: exact projection, for example with the pool-adjacent-violators algorithm (PAVA) [4], repairs the final output cheaply, but the intermediate predictions that guidance, early stopping, or a simulation monitor may read remain unconstrained, so each must be projected when it is read, and projecting inside the reverse process [11] also changes the input of the next denoising step (Figure 1). In both paradigms, validity is encouraged in training or imposed by an external repair rather than guaranteed by the generator's own decoder. We therefore ask whether every book that a diffusion model decodes, final or intermediate, can be valid by construction, at negligible computational cost and with the diffusion objective and sampler unchanged.

Answering this question raises three challenges. **CH1: Coupled constraints.** In raw price coordinates, whether a book is crossed depends jointly on two prices, and whether a ladder is ordered depends on every adjacent pair, so the price rules bound differences between outputs, not any single output. **CH2: Validity without repair.** Even in coordinates that isolate each constraint, a network can output a value below its bound, and validity must hold for every output without changing the diffusion objective, adding a projection to sampling, or collapsing the generated distribution. **CH3: Attribution and value.** Zero violations could instead come from easier coordinates or a stronger denoiser, and exact projection also yields valid final books, so the source of the guarantee must be isolated and its value beyond repair measured. The difficulty thus lies in guaranteeing validity for every decoded output with the denoiser, objective, and sampler unchanged, and in showing where the guarantee comes from and what it buys over repair.

We address these challenges with LegalLOB, which changes only the output interface (Section 3.4): the coordinates in which books are encoded for training and from which clean-state predictions are decoded (Figure 2). To decouple the constraints (CH1), its constraint-aligned representation **M1** (Section 4.1) describes each future book by its midpoint relative to the last observed midpoint, the bid–ask spread, the gaps between adjacent levels on each side, and the depths, so that each constraint becomes a lower bound on a single quantity while the midpoint stays free; M1 extends the midpoint, spread, and level differences of DiffLOB [36] by making every constraint explicit. To guarantee validity without repair (CH2), its **softplus decoder M2** (Section 4.2) keeps the spread, gaps, and depths above their bounds and rebuilds both price ladders by cumulative sums from the midpoint, so that in exact arithmetic every network output, final or intermediate, decodes to a strictly valid book (Proposition 4.1); as a smooth bijection onto the interior of the valid set, the decoder does not collapse outputs onto its boundary, as projection does. Building ordered values from positive increments is a standard reparameterization (Section 2); our contribution is the representation that extends it to a full two-sided book around a free midpoint. The name LegalLOB refers to this structural validity, not to legal compliance.

To attribute the guarantee and measure its value beyond repair (CH3), we compare LegalLOB with variants that share its denoiser architecture, hyperparameters, data, and training budget but differ in their output coordinates and in where validity is imposed: nowhere, in training, after sampling, or at every denoising step (Section 4.3). A coordinates-only variant (Coord-only), which keeps M1 with identity links, tests whether the coordinates alone suffice, and a recursive rollout that reads every decoded prediction shows when built-in validity matters. Across four Nasdaq datasets, this comparison shows that LegalLOB is the only learned model with no violations before repair, and the only one whose decoded predictions during a rollout can all be used without projection; post-hoc projection remains a strong baseline when only the final output matters. The guarantee costs little generation time, but the best repair-based baseline still leads on most fidelity metrics. In summary, we contribute:

- **A constraint-aligned representation** of complete L2 books in which each structural constraint is a lower bound on one coordinate;
- **A softplus decoder** that makes every decoded book valid before any repair;
- **A controlled comparison** that locates the source of the guarantee, shows when it matters beyond repairing the final output, and measures its cost.

2 Related Work

Event-level LOB generation. Event-level models generate order messages or state-conditioned order flow and update the book through a simulator or matching engine [5, 14–16, 22, 23, 26, 29]; validity then follows from the event vocabulary, action masks, or the matching engine. LegalLOB works at a different level: it generates continuous market-by-price trajectories, so invalid books must be prevented in the generated state itself.

Diffusion models for financial time series and LOBs. Diffusion, score-based, and flow-based models are flexible conditional generators for sequential data, from time series [2, 32] to user-behavior sequences in recommendation [7, 9]. For LOBs, DiffVolume generates depth profiles under given liquidity conditions [35], DiffLOB generates snapshot trajectories under trend, volatility, liquidity, and imbalance regimes [36], and Painting the Market treats L2 books as images and inpaints them [3]; none guarantees ordered prices and non-negative depth in every generated book. Financial prediction models capture temporal dependence [38] and higher-order interactions among stocks [1]; they forecast prices or returns, whereas LegalLOB constrains the generated book.

Constrained generation. Penalties, rejection, and guidance impose user-specified conditions in time-series generation [13], and guidance steers diffusion samples toward desired properties [8, 12]; penalties and guidance encourage validity without guaranteeing it, and rejection guarantees it only by discarding samples. Reflected, constrained-domain, mirror, and projected diffusion models and learned approximate mirror maps handle broader constraint sets [11, 17, 18, 24, 25] by modifying the stochastic dynamics, repeating a constraint operation during sampling, or learning an additional map. For ordered quantities, accumulating positive increments is a standard alternative, used for ordered vectors in probabilistic programming [10], non-crossing quantiles [39], and monotone networks [37]; LegalLOB applies it to complete L2 books and needs no constraint operator in the sampling loop.

Exact repair of ordered prices. Isotonic regression gives the least-squares projection onto an ordered set, and PAVA computes it exactly for a chain [4, 6]. PAVA is exact and inexpensive, and it is the natural choice when only the final book must be valid (Section 6).

3 Problem Formulation

3.1 Conditional Generation of L2 Books

A Level-2 snapshot with K price levels per side is the vector $\mathbf{x}_t = (\mathbf{a}_t, \mathbf{b}_t, \mathbf{v}_t^a, \mathbf{v}_t^b) \in \mathbb{R}^{4K}$, where the ask prices $\mathbf{a}_t = (a_t^{(1)}, \dots, a_t^{(K)})$ and the bid prices $\mathbf{b}_t = (b_t^{(1)}, \dots, b_t^{(K)})$ run from the best quote outward and $\mathbf{v}_t^a, \mathbf{v}_t^b$ are the corresponding displayed depths, all in the normalized units of Section 5.1. Given the history $\mathbf{H}_t = \mathbf{x}_{t-T_h+1:t}$, the task is to learn the conditional distribution $p_\theta(\mathbf{X}_t \mid \mathbf{H}_t)$ of the future block $\mathbf{X}_t = \mathbf{x}_{t+1:t+T_f}$.

3.2 Structural Validity

For a minimum spread $\delta > 0$, a minimum gap $\gamma \geq 0$ between adjacent price levels, and a depth floor $\epsilon_v \geq 0$, the set of valid L2 books is

$$\begin{aligned}\mathcal{F}_{\delta,\gamma,\epsilon_v} = \{\mathbf{x} : a^{(1)} - b^{(1)} \geq \delta, \quad v_i^q \geq \epsilon_v, \\ a^{(k)} - a^{(k-1)} \geq \gamma, \quad b^{(k-1)} - b^{(k)} \geq \gamma, \\ k = 2{:}K, \quad i = 1{:}K, \quad q \in \{a, b\}\}.\end{aligned} \tag{1}$$

A generated block is structurally valid when all of its T_f snapshots lie in $\mathcal{F}_{\delta,\gamma,\epsilon_v}$; its interior $\operatorname{int}\mathcal{F}_{\delta,\gamma,\epsilon_v}$ is the same set with every inequality strict. These constraints concern continuous market-by-price states. Rounding to the tick grid and to integer sizes is discussed in Section 4.4, whereas event-level properties such as order identity, queue priority, matching, trading halts, and price limits require an event model and are outside our scope.

3.3 Conditional Diffusion Models

Following denoising diffusion probabilistic models [21], let $\mathbf{z}_0$ denote the coordinates of the future block. At diffusion step n, $\mathbf{z}_n = \sqrt{\bar{\alpha}_n}\,\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_n}\,\boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and a history-conditioned denoiser with the DiffLOB architecture [36] predicts the added noise by minimizing

$$\mathcal{L}_{\text{diff}} = \mathbb{E}\,\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_n, n, \mathbf{H}_t)\|_2^2\,. \tag{2}$$

All variants sample with DDIM [30]. At each reverse step, the sampler forms a clean-state prediction $\widehat{\mathbf{z}}_0^{(n)}$; decoding it gives the book that a reader of intermediate states would see.

3.4 Output Interfaces and Validity Along the Sampling Path

An *output interface* consists of a coordinate map T, which encodes a future block as $\mathbf{z}_0 = T(\mathbf{X}_t)$, its inverse $D = T^{-1}$, which decodes model coordinates into books (up to the encoder clamp of Section 4.2), and, for repair-based interfaces, an exact projection $\Pi_{\mathcal{F}}$ of decoded books, applied once after sampling or at every reverse step (Section 4.3). A diffusion model is *valid along its sampling path* if every decoded clean-state prediction $D(\widehat{\mathbf{z}}_0^{(n)})$, $n = N, \dots, 1$, is structurally valid as delivered, that is, after the interface's own projection, if any; the final output is the last of these. A decoder whose image lies in $\mathcal{F}_{\delta,\gamma,\epsilon_v}$ gives this property without projection, for any denoiser and any noise (in exact arithmetic; Section 4.4 treats float32). A decoder with identity links can leave $\mathcal{F}_{\delta,\gamma,\epsilon_v}$, so each read needs its own projection: without feedback, this keeps the path at one call per read, whereas feeding the projected predictions back to the sampler can change it. We therefore seek an output interface whose decoder maps every point of the coordinate space into $\mathcal{F}_{\delta,\gamma,\epsilon_v}$ and that leaves the denoiser architecture, the objective in Eq. (2), and the DDIM sampler unchanged.

4 Method

LegalLOB is the conditional diffusion model of Section 3 with a new output interface (Section 3.4), built from the representation M1 (Section 4.1) and the softplus decoder M2 (Section 4.2). Section 4.3 describes the alternative interfaces against which it is compared. Figure 2 shows the diffusion model and the three interfaces.

4.1 Constraint-Aligned Book Representation

Predicting all price levels directly leaves their ordering implicit (CH1). M1 instead separates the free movement of the book from its constrained shape. Each history–future window uses the midpoint of the last history snapshot as a fixed anchor, $m_t^\star = \frac{1}{2}\big(a_t^{(1)} + b_t^{(1)}\big)$, and describes each future snapshot by

$$\begin{aligned}&\widetilde{m} = \tfrac{1}{2}(a^{(1)} + b^{(1)}) - m_t^\star, \qquad s = a^{(1)} - b^{(1)}, \\ &g_k^a = a^{(k)} - a^{(k-1)}, \qquad g_k^b = b^{(k-1)} - b^{(k)}, \qquad k = 2{:}K.\end{aligned} \tag{3}$$

The anchor removes the price level but keeps the midpoint path. The spread s, the ask gaps $\mathbf{g}^a$, the bid gaps $\mathbf{g}^b$, and the depths $\mathbf{v}^a, \mathbf{v}^b$ are exactly the quantities constrained in Eq. (1), so every ordering constraint becomes a lower bound on a single coordinate.

4.2 Softplus Decoder

M1 exposes the constrained quantities but does not keep their predicted values above their bounds (CH2). M2 therefore writes each constrained quantity as its lower bound plus a softplus term, so that validity is enforced by decoding, not encouraged by training.

Let $\rho(u) = \log(1 + e^u)$ be the softplus function, which maps every real number to a positive one. For training targets, M2 keeps $\widetilde{m}$ and maps the excesses $s-\delta$, $g^a-\gamma$, $g^b-\gamma$, and $v^q-\epsilon_v$ through ρ^{-1}, after clamping each excess at a small $\epsilon_{\text{enc}} > 0$ so that observations exactly on a bound stay finite; each coordinate is then standardized with statistics of the training windows. At generation time, M2 undoes the standardization and decodes the latent vector:

$$\begin{aligned}s &= \delta + \rho(u_s), & v_i^q &= \epsilon_v + \rho(u_i^{v,q}), \\ g_k^a &= \gamma + \rho(u_k^a), & g_k^b &= \gamma + \rho(u_k^b), \\ m &= m_t^\star + \widetilde{m}, & & \\ a^{(1)} &= m + s/2, & b^{(1)} &= m - s/2, \\ a^{(k)} &= a^{(1)} + \sum_{j=2}^{k} g_j^a, & b^{(k)} &= b^{(1)} - \sum_{j=2}^{k} g_j^b.\end{aligned} \tag{4}$$

Proposition 4.1 (Validity by construction). *For a fixed anchor $m_t^\star$, $\delta > 0$, $\gamma \geq 0$, and $\epsilon_v \geq 0$, the decoder $D_{m_t^\star} : \mathbb{R}^{4K} \to \operatorname{int}\mathcal{F}_{\delta,\gamma,\epsilon_v}$ defined by Eq. (4) is, in exact arithmetic, a smooth bijection*

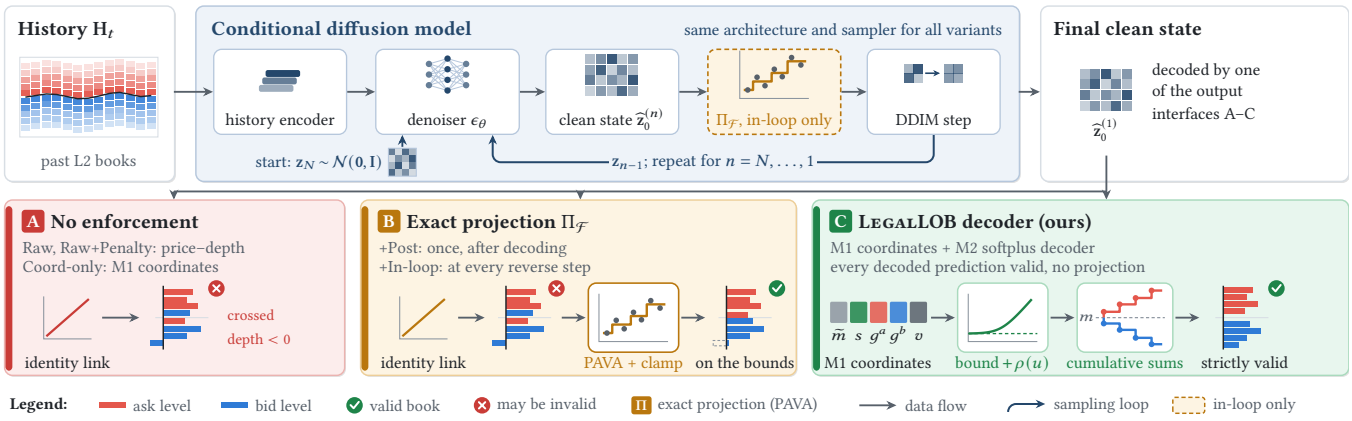


Figure 2: Framework. All variants share the denoiser architecture, the noise-prediction objective (Raw+Penalty adds a penalty), and the DDIM sampler (top), and each learned variant is trained separately. They differ in how the clean-state prediction is decoded into a book (bottom): (A) identity links; (B) identity links followed by exact projection, once after decoding or at every reverse step; (C) the LegalLOB softplus decoder.

Table 1: Compared output interfaces (✓: guaranteed; valid at every step: every decoded clean-state prediction is valid; path kept: the sampling path is unchanged).

Variant	Coordinates	Validity from	Valid output	Valid at every step	Path kept
Raw	price–depth	none	✗	✗	✓
Raw+Penalty	price–depth	training penalty	✗	✗	✓
Coord-only	M1	none	✗	✗	✓
Raw/Coord-only+Post	as parent	$\Pi_{\mathcal{F}}$ on output	✓	✗	✓
Raw/Coord-only+In-loop	as parent	$\Pi_{\mathcal{F}}$ at every step	✓	✓	✗
LegalLOB	M1	softplus decoder	✓	✓	✓

with inverse

$$\begin{aligned} \tilde{m} &= (a^{(1)} + b^{(1)})/2 - m_t^\star, \quad u_s = \rho^{-1}(a^{(1)} - b^{(1)} - \delta), \\ u_k^a &= \rho^{-1}(a^{(k)} - a^{(k-1)} - \gamma), \quad u_i^{v,q} = \rho^{-1}(v_i^q - \epsilon_v), \\ u_k^b &= \rho^{-1}(b^{(k-1)} - b^{(k)} - \gamma). \end{aligned} \tag{5}$$

Proof. Softplus is a smooth bijection from $\mathbb{R}$ to $(0, \infty)$, so the decoded spread, gaps, and depths lie strictly above their lower bounds, and the cumulative sums turn these bounds into the strict ordering constraints of Eq. (1). Conversely, every point of int $\mathcal{F}_{\delta,\gamma,\epsilon_v}$ has the unique finite preimage given by Eq. (5). □

The inverse in Eq. (5) encodes the training targets. The clamp ϵ_{enc} moves boundary observations into the interior, and standardization keeps the bijection since all scales are positive.

4.3 Alternative Output Interfaces

To locate the source of the guarantee (CH3), we compare LegalLOB with output interfaces that impose validity elsewhere or not at all (Table 1). Raw predicts prices and depths directly; Coord-only keeps the M1 coordinates but decodes them with identity links, which tests whether the coordinates alone yield valid books; Raw+Penalty follows the soft-constraint formulation of constrained time-series generation [13]; and exact projection is applied either once to the decoded output (+Post) or to the decoded clean-state prediction at every denoising step (+In-loop). For the increasing price chain $\mathbf{c} = (b^{(K)}, \ldots, b^{(1)}, a^{(1)}, \ldots, a^{(K)})$, let the offsets $\mathbf{o}$ start at zero and grow by γ between adjacent levels and by δ between the best bid and the best ask. The exact projection is $\mathbf{c}^{\text{proj}} = \text{PAVA}(\mathbf{c} - \mathbf{o}) + \mathbf{o}$, with depths clamped at $\epsilon_v = 0$; this is the Euclidean projection onto the ordered price chain [4, 6].

In-loop projection changes the clean-state prediction that the next DDIM step receives, post-hoc projection does not, and LegalLOB needs neither (Algorithm 1). Raw+Penalty adds $\lambda \mathcal{L}_{\text{pen}}$ to $\mathcal{L}_{\text{diff}}$, where $\mathcal{L}_{\text{pen}}$ is the sum of the mean squared shortfalls of the spread, the ask gaps, the bid gaps, and the depths of the decoded clean-state prediction below their bounds. We use a single weight, $\lambda = 10^{-5}$, to test whether this finite training pressure gives the same hard guarantee, not to map the trade-off.

Algorithm 1 DDIM sampling with the three output interfaces

Require: history $\mathbf{H}_t$; variant r with coordinate map T_r and decoder $D_r = T_r^{-1}$; denoiser ϵ_θ; steps N

$\mathbf{z}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $n = N, \ldots, 1$ **do**
  $\hat{\mathbf{z}}_0^{(n)} \leftarrow \text{PredictClean}(\mathbf{z}_n, \epsilon_\theta(\mathbf{z}_n, n, \mathbf{H}_t), n)$
  if r is an in-loop variant **then**
    $\hat{\mathbf{z}}_0^{(n)} \leftarrow T_r(\Pi_{\mathcal{F}}(D_r(\hat{\mathbf{z}}_0^{(n)})))$
  end if
  $\mathbf{z}_{n-1} \leftarrow \text{DDIMStep}(\mathbf{z}_n, \hat{\mathbf{z}}_0^{(n)}, n)$
end for
$\hat{\mathbf{X}}_{\text{pre}} \leftarrow D_r(\hat{\mathbf{z}}_0^{(1)})$
if r is a post-hoc variant **then**
  $\hat{\mathbf{X}}_{\text{out}} \leftarrow \Pi_{\mathcal{F}}(\hat{\mathbf{X}}_{\text{pre}})$
else
  $\hat{\mathbf{X}}_{\text{out}} \leftarrow \hat{\mathbf{X}}_{\text{pre}}$
end if
return $\hat{\mathbf{X}}_{\text{pre}}$ (before repair) and $\hat{\mathbf{X}}_{\text{out}}$ (final output)

Table 2: Datasets and shared configuration (chronological splits; statistics fitted on the training split).

Dataset	Sampling	Split	Snapshots per ticker
Fixed-grid 1-Hz panel (Public TotalView-ITCH)			
AAPL/AMZN/GOOG-1Hz	last valid book per second	3/1/2 days	140,400
Event-time panel (Databento XNAS.ITCH MBP-10)			
GOOG-Event	every valid book update	70/15/15 + purge	856,984

Preprocessing		*Shared model configuration*	
Price unit	10 exchange ticks	Backbone	128 ch., 16 res. blocks
Bounds	δ=γ=0.1	Diffusion	VP-DDPM, 100 levels
Depth transform	p99 clip, $\sqrt{\cdot}$, c_v=15	Optimizer	Adam, 10^{-4}
Floor / enc. clamp	ϵ_v=0 / ϵ_{enc}=10^{-12}	Batch / epochs	128 / $\leq$50
History / future	50 / 10 snapshots	EMA	0.999
Anchor	last history midpoint	Sampling	DDIM-20

4.4 Cost, Precision, and Discretization

For a block of T_f snapshots with K levels, M2 needs $O(T_f K)$ operations (links and cumulative sums), as does one exact PAVA projection; LegalLOB adds nothing to the sampling loop.

Proposition 4.1 holds in exact arithmetic. In float32, the decoder rebuilds prices one level at a time and, whenever rounding would make a spread or gap exactly equal to its bound, moves the price to the next representable value; this local guard implements the strict inequality of the decoder and is not a projection. Softplus depths can underflow to the depth floor for extreme inputs, and recursive rollouts re-anchor each new block on the previous one, so we check validity with an explicit numerical tolerance (Section 5.3).

Discretization. Exchange prices lie on a tick grid and sizes are integers. With $\delta = \gamma$ equal to one tick (Table 2), a decoded book stays valid on the grid if each price p, measured in ticks, is rounded to $\lfloor p+\frac{1}{2} \rfloor$ (ties rounded up). This map is non-decreasing and satisfies $\lfloor p + 1 + \frac{1}{2} \rfloor = \lfloor p + \frac{1}{2} \rfloor + 1$, so spreads and gaps of at least one tick remain at least one tick; round-half-to-even, the default in many numerical libraries, lacks this property and can merge two levels one tick apart. Rounding depths to whole shares keeps them non-negative. Rounding cannot uncross a crossed book, so it is safe only for valid books; all reported results use the continuous outputs.

5 Experiments

The experiments answer three research questions. **RQ1:** How frequent and how large are structural violations, and which component provides the guarantee (Section 5.4)? **RQ2:** When does built-in validity matter beyond exact repair (Section 5.5)? **RQ3:** What does the guarantee cost in fidelity and computation (Sections 5.6 and 5.7)? RQ1 and RQ2 answer CH3; RQ3 measures the price of the guarantee.

5.1 Datasets and Preprocessing

Following prior L2 generation studies, we use the top ten price levels and fixed-length history–future windows [3, 35, 36]. The fixed-grid panel (Table 2) keeps the last valid regular-hours book in each second for AAPL, AMZN, and GOOG on six trading days in 2019: Jan. 30, Mar. 27, and Jul. 30 for training, Aug. 30 for validation, and Oct. 30 and Dec. 30 for testing. The event-time panel keeps every valid GOOG book update on Jan. 30, 2019, split chronologically 70/15/15 with 100-row purged gaps around both split points; this denser and irregularly timed sequence stress-tests the method. Models are trained separately for each dataset, so the evidence covers several settings, not generalization to unseen assets. Prices are expressed in units of ten ticks, and depth, following related LOB diffusion models [35, 36], is clipped at the 99th percentile of the training split, square-root transformed, and scaled by c_v. Validity is checked in these normalized units; because the depth transform is increasing and maps zero to zero, non-negative depth means the same in shares. Financial functionals are computed after mapping prices back to ticks and depths back to clipped shares. All preprocessing statistics and thresholds are fitted on the training split, and no window spans two trading days or a data gap.

5.2 Compared Variants and Implementation

The learned variants (Raw, Raw+Penalty, Coord-only, and LegalLOB) use the same DiffLOB denoiser architecture, with temporal attention and gated WaveNet-style residual blocks [33, 34, 36], the same hyperparameters, data, and training budget, and are trained separately for each dataset and seed. The projection variants (+Post and +In-loop) reuse the checkpoint of their parent, Raw or Coord-only. Tables 6 and 7 compare the post-hoc variants with LegalLOB; the in-loop variants enter the rollout study (Tables 4 and 5) and, for Raw, the cost study (Table 8). Persistence repeats the last history snapshot. Paired comparisons share histories, Gaussian noise, and the DDIM-20 schedule; the optional action conditioning of the backbone is disabled throughout. PAVA always acts on decoded books in normalized price units, never on standardized model coordinates.

Training and hyperparameter selection. Each learned variant minimizes Eq. (2) (plus $\lambda \mathcal{L}_{\text{pen}}$ for Raw+Penalty) with Adam. We keep the checkpoint with the lowest validation loss under fixed noise, stop after ten epochs without improvement or at 50 epochs (every run reached the 50-epoch limit), and evaluate the exponential moving average (EMA) of its weights. No hyperparameter is tuned per variant. The bounds follow market structure rather than tuning: $\delta = \gamma = 0.1$ is one exchange tick in the 10-tick price unit, the smallest spread and level gap that a valid book can show, and $\epsilon_v = 0$ encodes non-negative depth; ϵ_{enc} only sets the latent value of observations that lie exactly on a bound (Section 6). The backbone depth, number of noise levels, learning rate, batch size, EMA, and depth transform follow DiffLOB [36]; the 50/10 window and DDIM-20 sampling are our choices, and the penalty weight is a single operating point (Section 4.3).

5.3 Evaluation Metrics

Validity. A snapshot is invalid when its spread, any adjacent gap, or any depth violates Eq. (1) by more than a tolerance of 2×10^{-6}. Each evaluation covers 6,400 histories with ten future snapshots each. We report the share of invalid snapshots and, as the size of the repair, the 95th percentile (p95) of the absolute price change that PAVA makes on the levels it modifies.

Fidelity. Point MSE averages the squared error over the normalized coordinates of each snapshot; projected variants are scored on the output a user would receive. Temporal diagnostics are the

Table 3: Structural violations before repair: invalid snapshots (%; mean ± SD over three checkpoints; variants in Table 1).

Variant	AAPL-1Hz	AMZN-1Hz	GOOG-Event	GOOG-1Hz
LegalLOB (M1 + M2)	**0**	**0**	**0**	**0**
Coord-only (M1)	99.9 ±0.08	48.3 ±0.90	78.5 ±0.36	59.3 ±0.59
Raw	100.0 ±0.002	59.9 ±1.03	84.9 ±0.43	70.1 ±0.72
Raw+Penalty	100.0 ±0.0003	59.6 ±0.41	84.4 ±1.28	69.4 ±0.85

Wasserstein-1 (W1) distance of returns, the integrated absolute error (IAE) of the autocorrelation functions (ACF) of absolute returns and spreads at lags 1, 2, and 5, and the error of cross-level volume correlations. The recursive rollout generates ten consecutive blocks and compares spread and volume drift, return volatility, and temporal correlation with the real paths. The projection correction of a block is its normalized displacement $\|\mathbf{X} - \Pi_{\mathcal{F}}(\mathbf{X})\|_2$, and the cumulative repair sums these corrections along a path, measuring how much a method intervenes.

Financial functionals. We compare generated and real distributions of the microprice offset from the midpoint, the ten-level depth imbalance (OI10), the displayed liquidity Liq@k, and the slippage of a market order of fixed size Q, set to the median positive best-level depth of the real evaluation data. We define Liq@k = $(L_k^a + L_k^b)/(2Q)$, where L_k^a is the displayed ask depth within k ticks of the best ask and L_k^b the bid depth within k ticks of the best bid; slippage is the gap between the order's average fill price and the best quote, defined when the ten displayed levels can fill Q. Following LOB-Bench [27], we use snapshot-level statistics only (not its message-level ones); they measure snapshot sensitivity rather than queue-aware execution or backtest performance [20].

Intermediate-state consumer. On GOOG-Event, a fail-closed risk monitor reads the decoded clean-state prediction at every denoising step and each block endpoint of a 100-step rollout. It stands in for pipelines that use states before the final output, such as guidance or early stopping (Section 6). A request is accepted only if all its snapshots are finite and valid. For accepted snapshots, the monitor raises a risk alarm when a buy or sell order of size Q cannot be filled, when the larger of the two slippages exceeds its 90th percentile, or when Liq@5 falls below its 10th percentile; here Q and both percentiles are computed on the training split. LegalLOB is read directly, Coord-only+Post repairs only block endpoints, and Coord-only+In-loop projects every decoded prediction and continues sampling from the projected state.

Probabilistic scores. The conditional evaluation uses 512 common GOOG-Event histories and $M = 32$ paired trajectories per checkpoint. With $d_{\mathrm{RMS}}(u, v) = \|u - v\|_2/\sqrt{400}$, the energy score is

$$\mathrm{ES} = M^{-1}\sum_{m} d_{\mathrm{RMS}}(X_m, y) - (2M^2)^{-1}\sum_{m,m'} d_{\mathrm{RMS}}(X_m, X_{m'}). \quad (6)$$

We also report the coordinate-averaged continuous ranked probability score (CRPS) [19], a variogram score of order 1/2 over midpoint increments, spreads, and depth imbalances [28], the coverage of central 50% and 90% intervals, and the pairwise RMSE between samples as a measure of dispersion.

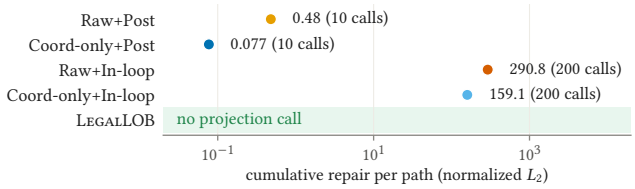


Figure 3: Cumulative repair per 100-step GOOG-Event path (normalized L_2, log scale; mean ± SD over three checkpoints, Table 4), with the number of projection calls per path.

Statistics and reproducibility. Three independently trained checkpoints (seeds 0, 1, and 2) each use three paired sampling seeds on common histories. We average over sampling seeds within a checkpoint and report the mean ± sample standard deviation (SD) across checkpoints. Persistence is deterministic. Exact zeros, undefined values, and inapplicable quantities are marked explicitly (0, undef., N/A), and bold and underline mark the best and second-best stochastic method with valid outputs. The repository provides the preprocessing, training, and evaluation scripts for the four settings.

5.4 RQ1: Violations and the Source of the Guarantee

Structural violations are frequent, and only the softplus links remove them. Table 3 removes one component at a time: Coord-only drops the softplus links, and Raw also drops the constraint-aligned coordinates. Coord-only, Raw, and Raw+Penalty produce invalid snapshots at every checkpoint, in 48–100% of cases on average, whereas LegalLOB produces none. The coordinates alone therefore do not provide the guarantee; the softplus links of M2 do. Raw+Penalty is almost identical to Raw, so the chosen weight exerts little pressure. A finite penalty gives no guarantee by construction, but because we test a single weight, Table 3 cannot rule out that a larger weight would approach zero violations.

M1 shrinks violations, but even small ones matter. A paired audit of all evaluated snapshots shows that PAVA changes about half to all of the parent snapshots; the p95 price correction reaches about 1.8 ticks with raw coordinates but stays below 0.2 ticks with the M1 coordinates, and corrections above one tick occur mostly with raw coordinates. Even small violations matter, because validity is binary: a spread or gap below one tick cannot occur on the exchange's price grid, and a fail-closed consumer rejects such a book. The audit also confirms that PAVA is a strong endpoint baseline: it removes all violations, never increases the normalized MSE, and agrees with an independent float64 computation up to round-off. PAVA reaches the boundary of $\mathcal{F}$ after decoding, whereas LegalLOB generates points strictly inside it (Figure 2, B and C).

5.5 RQ2: Intermediate States in Recursive Rollouts

Table 4 and Figure 3 report a 100-step GOOG-Event rollout over 640 histories, three sampling seeds, and three checkpoints. Post-hoc PAVA is called 10 times per path, once at each block endpoint, whereas in-loop PAVA is called 200 times, once at each denoising

Table 4: 100-step GOOG-Event rollout (mean ± SD over three checkpoints; corrections are normalized L_2 distances; bold and underline rank the methods with valid outputs; N/A: not applicable).

	Validity and repair burden				*Rollout fidelity*			
Method	Proj. calls per path ↓	Invalid (%) at h=100 ↓	Correction per call ↓	Cumulative repair ↓	Spread drift ↓	Volume drift ↓	Return-σ drift ↓	Temporal corr. ↑
Raw	0	53.21 ±0.42	N/A	N/A	0.071 ±0.007	0.112 ±0.003	0.200 ±0.004	0.020 ±0.003
Raw+Post	10	**0**	0.048 ±0.001	0.48 ±0.01	0.071 ±0.007	0.112 ±0.003	0.203 ±0.004	0.020 ±0.004
Raw+In-loop	200	**0**	1.454 ±0.096	290.8 ±19.3	0.084 ±0.006	0.101 ±0.003	**0.188** ±0.002	0.021 ±0.005
Coord-only	0	41.94 ±3.65	N/A	N/A	0.063 ±0.012	0.067 ±0.003	0.293 ±0.016	0.012 ±0.003
Coord-only+Post	10	**0**	**0.0077** ±0.0003	**0.077** ±0.003	0.062 ±0.013	0.066 ±0.003	0.295 ±0.016	0.013 ±0.004
Coord-only+In-loop	200	**0**	0.796 ±0.038	159.1 ±7.6	0.062 ±0.011	**0.065** ±0.003	0.277 ±0.017	0.005 ±0.007
LegalLOB	**0**	**0**	N/A	N/A	**0.059** ±0.002	0.091 ±0.004	0.264 ±0.013	**0.033** ±0.006

Table 5: Rejected intermediate requests in the GOOG-Event rollout (%; lower is better; mean ± SD over three checkpoints; as delivered: after the method's own projection, if any).

	Before projection		*As delivered*	
Method	Snapshots	Blocks	Snapshots	Paths with a rejection
LegalLOB	**0**	**0**	**0**	**0**
Coord-only+Post	51.9 ±1.0	84.6 ±0.9	51.9 ±1.0	100
Coord-only+In-loop	50.4 ±0.8	83.5 ±0.8	$(1.0 \pm 0.3) \times 10^{-4}$	0.19 ±0.06

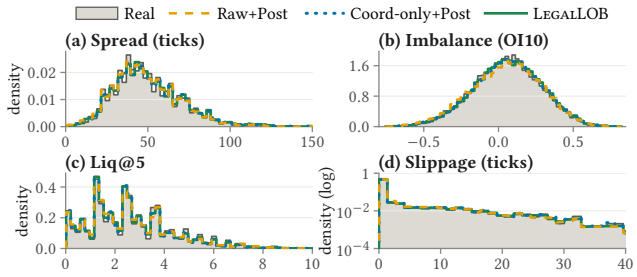


Figure 4: GOOG-Event marginal distributions of real data (gray) and valid outputs; Table 6 gives W1 for (b)–(d) on all datasets. Panel (d) is cut at 40 ticks (less than 1% of the mass lies beyond this point).

step, and changes the states inside the sampler. Each in-loop call also moves the state much further than a post-hoc call (about 30 times further with the Raw model), so the larger cumulative repair is not explained by the number of calls alone.

All valid-output methods end with no violations, but only LegalLOB needs no projection along the way: every intermediate request to it is accepted without repair (Table 5). Coord-only+Post leaves about half of the snapshot requests unavailable, and Coord-only+In-loop makes almost all of them available only through 200 projections per path and a large cumulative repair; its 11 remaining rejections among 11.5 million requests are float32 rounding cases at the anchor that pass in float64. Projection inside the sampler also changes the generated path: after 100 steps, the in-loop and post-hoc versions of Coord-only reach different books, and the monitor's alarms disagree on 2.4 ± 0.3% of their delivered snapshots. The extra calls buy availability rather than fidelity, since the rollout fidelity of Coord-only+In-loop is close to that of Coord-only+Post.

5.6 RQ3: The Cost in Fidelity

In Table 6, LegalLOB has the lowest microprice W1 among the stochastic methods on three datasets and the lowest Liq@5 W1 on both GOOG datasets, while the best method on temporal diagnostics varies by asset and statistic. On GOOG-Event, the generated and real marginal distributions closely overlap (Figure 4), and every stochastic method matches the real probability of zero slippage to within one percentage point; the guarantee therefore does not collapse the marginal distributions. It is not free in fidelity, however: Coord-only+Post outperforms LegalLOB on two thirds of the metric–dataset pairs, by a wide margin on spread ACF for AAPL-1Hz and GOOG-1Hz and on absolute-return ACF for AMZN-1Hz, and LegalLOB has the highest point MSE among the stochastic methods on the three 1-Hz datasets, most clearly on AMZN-1Hz. PAVA itself changes a snapshot's microprice and slippage by at most a few hundredths of a tick on average, but its Liq@5 by up to about 0.5, while leaving the full-fill rate almost unchanged.

Why Persistence wins on point error. Persistence beats every generative model on point MSE and on all four snapshot functionals, yet loses on every defined temporal diagnostic. Over ten one-second or event-time steps the book changes little, so the last book is close to the conditional mean and is itself a real book. A random sample, by contrast, incurs twice the minimum squared error: for independent X and Y with the same conditional law, $\mathbb{E}\|X - Y\|_2^2$ is twice the MSE of the conditional mean. Point MSE and snapshot marginals therefore cannot separate a useful generator from copying the past, whereas temporal diagnostics and proper scores can; Persistence's constant midpoint also leaves its absolute-return ACF undefined.

LegalLOB's probabilistic scores stay close to the best repair-based baseline. On GOOG-Event (Table 7), LegalLOB has the most dispersed ensemble, beats the deterministic Persistence baseline on all three proper scores, and stays within a few percent of Coord-only+Post. Neither is well calibrated: both central 50% intervals over-cover (about 0.60), and LegalLOB's 90% interval (about 0.88), although the closest to its target, slightly under-covers.

5.7 RQ3: The Cost in Computation

Table 8 times the seed-0 checkpoints for the three 1-Hz datasets on one Apple M3 Ultra workstation with PyTorch 2.8 (MPS backend). The end-to-end time includes preprocessing, DDIM-20 sampling, decoding, data transfer, and any repair. LegalLOB keeps 99.6% of

Table 6: Fidelity of final outputs (lower is better; mean ± SD over three checkpoints; bold and underline: best and second-best stochastic method; gray: deterministic Persistence).

			Snapshot financial functionals				*Single-window temporal diagnostics*			
Dataset	Method	Point MSE	Microprice W1	Imbalance W1 ($\times 10^{-2}$)	Slippage W1	Liq@5 W1	Return W1 ($\times 10^{-3}$)	$\lvert r \rvert$ ACF IAE	Spread ACF IAE	Volume corr. MAE ($\times 10^{-2}$)
AAPL-1Hz	Persistence	0.2642	0.013	0.2	0.006	0.351	9.7	undef.	1.4268	8.76
	Raw+Post	**0.4078** ±0.0071	0.119 ±0.016	2.7 ±0.3	0.089 ±0.007	0.508 ±0.042	**3.2** ±0.7	0.1988 ±0.0267	0.4500 ±0.0533	**1.53** ±0.02
	Coord-only+Post	0.4113 ±0.0044	**0.067** ±0.001	**1.6** ±0.3	0.079 ±0.002	**0.484** ±0.054	4.3 ±0.1	**0.0639** ±0.0181	**0.1972** ±0.0028	1.94 ±0.05
	LegalLOB	0.4158 ±0.0259	0.104 ±0.017	2.0 ±0.1	**0.068** ±0.004	0.804 ±0.084	5.8 ±0.1	0.0698 ±0.0032	1.3696 ±0.0332	1.91 ±0.04
AMZN-1Hz	Persistence	4.0165	0.173	0.1	0.044	0.019	37.6	undef.	0.5742	9.30
	Raw+Post	**6.5483** ±0.0503	3.746 ±0.099	2.2 ±0.3	1.767 ±0.034	1.151 ±0.028	**13.8** ±0.3	0.2240 ±0.0028	0.2011 ±0.0239	**1.20** ±0.02
	Coord-only+Post	9.4691 ±0.2255	1.731 ±0.050	**1.2** ±0.2	**1.239** ±0.047	**0.666** ±0.020	20.7 ±0.8	**0.0552** ±0.0122	0.0701 ±0.0074	1.26 ±0.02
	LegalLOB	10.8342 ±0.1823	**1.499** ±0.126	**1.2** ±0.1	1.449 ±0.107	0.762 ±0.052	24.4 ±0.5	0.2128 ±0.0217	**0.0680** ±0.0122	1.23 ±0.02
GOOG-Event	Persistence	0.2829	0.056	0.2	0.039	0.010	2.6	undef.	0.0773	9.50
	Raw+Post	1.0683 ±0.0227	0.657 ±0.042	1.2 ±0.2	0.303 ±0.001	0.124 ±0.002	**0.7** ±0.1	**0.0701** ±0.0106	**0.0076** ±0.0009	1.02 ±0.03
	Coord-only+Post	**0.4390** ±0.0009	0.232 ±0.015	**0.64** ±0.03	**0.073** ±0.002	0.100 ±0.002	0.92 ±0.02	0.0715 ±0.0134	0.0284 ±0.0018	**0.87** ±0.03
	LegalLOB	0.4705 ±0.0020	**0.224** ±0.011	0.8 ±0.1	0.080 ±0.009	**0.084** ±0.004	1.01 ±0.05	0.0828 ±0.0080	0.0368 ±0.0009	0.89 ±0.01
GOOG-1Hz	Persistence	1.9005	0.105	0.2	0.029	0.047	19.7	undef.	0.3744	8.98
	Raw+Post	**3.3638** ±0.0178	2.868 ±0.185	1.9 ±0.1	0.664 ±0.021	1.700 ±0.017	**6.2** ±0.1	0.0725 ±0.0022	0.0336 ±0.0080	1.36 ±0.04
	Coord-only+Post	3.8576 ±0.0204	2.092 ±0.114	**1.3** ±0.1	**0.244** ±0.028	1.333 ±0.025	7.6 ±0.5	**0.0643** ±0.0054	**0.0077** ±0.0037	**1.15** ±0.01
	LegalLOB	3.8786 ±0.0414	**2.064** ±0.062	**1.3** ±0.1	0.530 ±0.026	**1.220** ±0.012	7.0 ±0.3	0.0792 ±0.0036	0.0337 ±0.0027	1.20 ±0.01

Table 7: Probabilistic scores on GOOG-Event (mean ± SD over three checkpoints; coverage ranked by closeness to its target; gray: deterministic Persistence).

Metric	Persistence	Raw+Post	Coord-only +Post	**LegalLOB**
Energy score ↓	0.4567	0.5867 ±0.0046	**0.3095** ±0.0004	0.3157 ±0.0003
Marginal CRPS ↓	0.1832	0.2768 ±0.0020	**0.1452** ±0.0004	0.1513 ±0.0002
Variogram score ↓	0.1128	0.0874 ±0.0008	**0.0826** ±0.0002	0.0832 ±0.0001
50% cov. (target 0.50)	N/A	**0.456** ±0.003	0.604 ±0.002	0.606 ±0.004
90% cov. (target 0.90)	N/A	0.767 ±0.003	0.877 ±0.001	**0.8806** ±0.0003
Pairwise RMSE	0	0.4595 ±0.0015	0.5512 ±0.0012	0.5718 ±0.0040

Table 8: Generation cost for 1,280 future snapshots on the 1-Hz datasets (Raw and Coord-only are unprojected references; projection variants reuse their parent's training).

Method	Proj. calls per block	Time (s) ↓	Snapshots/s ↑	Training s/epoch ↓
Raw	0	0.9846 ±0.0011	1300.0	110.3 ±2.7
Raw+Post	1	1.0340 ±0.0068	1237.9	as parent
Raw+In-loop	20	1.8043 ±0.0161	709.4	as parent
Coord-only	0	0.9861 ±0.0013	1298.1	111.1 ±3.3
Coord-only+Post	1	1.0341 ±0.0063	1237.8	as parent
LegalLOB	**0**	**0.9885** ±0.0014	**1294.9**	111.2 ±2.4

the throughput of Raw without any projection, post-hoc PAVA adds about 5% to the runtime, and in-loop projection more than 80%. Coord-only+In-loop is not part of this benchmark; in the GOOG-Event rollout audit, which times it under the same conditions as LegalLOB, it generated about half as many snapshots per second.

6 Discussion and Conclusion

When to prefer LegalLOB over post-hoc PAVA. If only the final book is used, post-hoc PAVA is the natural default: it is exact, adds about 5% to the runtime, never increases point MSE, and, applied to Coord-only, outperforms LegalLOB on most fidelity metrics. LegalLOB matters when earlier states are read, for example by guidance that scores the clean-state prediction at every denoising step [12], by anytime sampling that stops early under a latency budget, or by monitors of recursive simulations. Projection then needs a call for every read (200 per 100-step rollout), and feeding the repaired states back to the sampler changes the generated path (Section 5.5). LegalLOB predictions are valid as generated, need no repair before use, and pass guidance gradients through a smooth decoder rather than through piecewise-linear PAVA. Our study covers direct use, endpoint repair, and repair fed back into sampling, but not projection on every read without feedback, which keeps the path at the cost of one call per read; a guidance experiment is left for future work.

Limitations. Our data cover three large-cap Nasdaq stocks in 2019, with three training days per 1-Hz dataset and one day for the event-time panel; large-tick, illiquid, other-venue, and stressed-market books may behave differently. The guarantee is analytic, but the violation rates of the baselines and the fidelity cost may change in other markets. We compare output interfaces on one backbone rather than against external LOB generators, and we evaluate a single penalty weight. In the real test data, 19–94% of adjacent gaps, depending on the dataset, sit exactly at the one-tick minimum, and the strict-interior decoder represents these boundary observations only through the encoder clamp; a model with an explicit point mass on the boundary is a natural extension.

Conclusion. LegalLOB makes continuous L2 validity a property of every book the generator decodes along its sampling path, through a simple reparameterization: M1 exposes the constrained quantities, M2 keeps every decoded prediction valid, and the controlled comparison shows that the softplus links, not the coordinates or the network, provide the guarantee. Exact PAVA remains an effective repair for final outputs and, applied to Coord-only, leads on most fidelity metrics, whereas LegalLOB is most useful when intermediate or recursive states are consumed: it removes repair at negligible computational cost and keeps a stochastic ensemble.

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